Week 2

Functions, Continuity and Differentiability Pointers

- For a real-valued function, co-domain is $\mathbb{R}$.
- A function is said to be continuous in $\mathbb{R}$ if it is continuous at all points in $\mathbb{R}$.
- If a function is differentiable at a point, then it is also continuous at that point.
- If a function is not differentiable at a point, then it may or may not be continuous at that point.
- If a function is not continuous at a point, then it is not differentiable at that point.
- A function $f(x)$ is continuous at a point x_i if for any sequence of x_i converging to x_0, $f(x_i)$ converges to $f(x_0)$.
- Higher order approximations are more accurate than linear approximation.
- The plot of linear approximation of a function f at a point v is tangent to f at the point $(v, (f(v))$.

Vectors Pointers

- If the vectors u and v belong to a vector space V, then their linear combination belongs to V
- Two vectors x and y are perpendicular to each other if
    - $x.y = 0$
    - $x^T y = 0$
- $[0, 0, 0]$ is perpendicular to any given vector.

- A hyperplane normal to a vector w with offset value b is given by $x : w^T x = b$.
- A line through a point u along vector v is given by $x : x = u + \alpha v$.
- $x^T y = x.y = \sum_{i=1}^{d} x_i y_i$

Length of a Vector $u(x, y)$

$\|u\| = \sqrt{x^2 + y^2}$

Linear approximation of univariate functions

The linear approximation of a function $f(x)$ at point a is given as:

Linear approximation of multivariate functions

The linear approximation of a function f of two variables x and y in the neighbourhood of (a, b) is:

$$L(x, y) = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b)$$

Gradient of a function

- Gradient of a function evaluated at a point is a vector.
- It points in the direction of the steepest ascent.
- Gradient of a function at a point is perpendicular to the contour through the point.

$$\nabla f(x, y) = [\frac{\partial f(x, y)}{\partial x}, \frac{\partial f(x, y)}{\partial y}]$$

Finding Directional Derivative

Finding directional derivative of a function $f(x, y)$ at a point (a, b) in the direction of the vector $< u, v >$

1. Find $\nabla f(x, y)$.
2. Find $\nabla f(a, b)$
3. Convert $< u, v >$ to unit vector.
4. Directional Derivative = Dot product of $\nabla f(a, b)$ and $< u, v >$.

Direction of steepest accent

$$\frac{\nabla f(a,b)}{\|\nabla f\|}$$

Equation of a line passing through points a and b

$$[x,y,z] = a + \alpha(b-a)$$

or

$$[x,y,z] = b + \alpha(a-b)$$